AI CLASS · PROJECT PROPOSAL · PRESENTATION OUTLINE

AI Model *Lab*

Click it. Test it. See the result.

One page per slide. Each page says **what is on screen**, **what the banner video shows** and **the points to cover**, so each presenter can write their own script in their own words. There are lined spaces at the bottom of each page for notes.

#	SLIDE	PROFESSOR'S SECTION	PRESENTER	SUGGESTED TIME
0	Intro video (21 s, with sound)	—	A starts it	0:30
1	AI Model Lab · title and team	§1 Topic & team	A	0:45
2	What the lab does	§2 Scope / functions	A	1:00
3	How the models learn	§3 Techniques	B	1:30
4	How they predict, and how we measure	§4 Techniques	B	1:30
5	Sea creature detector	§5 Result / UI prototype	C	1:15
6	Sound detective	§6 Result / UI prototype	C	1:15
7	RAG assistants (in progress)	Extra	D	1:00
8	What's coming	Extra	D	0:45
9	What we built · references · live site	§7 Conclusion & references	E	1:00

Times are a suggestion (about 10–11 minutes in total). Change them to fit your slot.

PRESENTERS

PRESENTER	ROLE	SLIDES	NAME
A	The lab	0–2	
B	The making	3–4	
C	The models	5–6	
D	What's next	7–8	
E	Conclusion	9	

CONTROLS

Open `/present` on the site → press **Space** (or the start button) → the intro plays and holds on its last frame → **Space / ↓ / → / PageDown** for the next slide, **↑ / ← / PageUp** to go back, **Home / End** for first / last, **F** for full screen. A presentation clicker works too.

One rule for every script: only say numbers that are on the slides or in this outline. They are all measured. If someone asks for something we haven't measured, say "we haven't measured that yet".

SLIDE 0 · INTRO VIDEO

Presenter A | About 0:30 | No slide text

Intro *video*

ON SCREEN

> The 21-second AI Model Lab intro with music. It flies through a swarm of small models, a photo is tested, the result appears, then the report card is stamped HOW IT FAILS. It stops on its last frame and waits.

POINTS TO COVER

- Before pressing Space: one sentence to the class, for example a greeting and "this is our project, AI Model Lab".
- Let the video play. Don't talk over it; the music carries it.
- When it stops, press Space and move straight into slide 1.

SLIDE 1 · §1 TOPIC & TEAM

Presenter A · The lab | About 0:45 | Banner: slanted, right half

AI Model *Lab*

ON SCREEN

AI CLASS · PROJECT PROPOSAL

AI Model *Lab*

- *Click it. Test it. See the result.*
- Presenter A · The lab
- Presenter B · The making
- Presenter C · The models
- Presenter D · What's next
- Presenter E · Conclusion

BANNER SHOWS

> A swarm of specimen boxes, each one a small AI model. One box, "No. 01 · Sea creature detector", glows in front. (Stamped SAMPLE: an illustration.)

POINTS TO COVER

- The topic in one line: a library of small AI models that anyone can try in a web browser.
- The idea behind it: each model does **one job**, and each one has an **honest report card** that shows how it fails, not just a score.
- Introduce the team: say each name and what part they will present.

Hand-off: stay on. Presenter A continues with slide 2.

SLIDE 2 · §2 SCOPE / FUNCTIONS

Presenter A · The lab | About 1:00 | Banner: floating monitor

What the lab *does*

ON SCREEN

What the lab *does*

- Try a small AI model right in your browser
- Use a sample, your own photo or sound, or live listening
- See an honest report card: the score and how it fails
- Nothing you add is uploaded: it runs on your device
- 2 models live, more planned

BANNER SHOWS

> A grid of nine model plates; a red highlight moves from one to the next: "one job each". (SAMPLE.)

POINTS TO COVER

- Walk through the five functions as a list; this is the professor's "scope / functions" slide.
- Two models are live today: the **Sea creature detector** (finds sea animals in a photo) and the **Sound detective** (finds everyday sounds in a recording). Others, such as Fresh or spoiled, are planned.
- Privacy: the model downloads to your device and runs there, so the photo or sound never leaves it.

Hand-off to Presenter B: "B will now explain how we actually made these models."

SLIDE 3 · §3 TECHNIQUES

Presenter B · The making | About 1:30 | Banner: band along the bottom

How the models *learn*

ON SCREEN

How the models *learn*

- Supervised learning on labelled data: 7,637 sea photos (Open Images, Aquarium Combined) and 13,825 sound clips (FSD50K, Freesound)
- Transfer learning: fine-tuned pretrained CNNs, YOLO26 Nano for detection and EfficientAT mn10 (AudioSet) for sound
- Train / validation / test split; the test set stays frozen: 767 photos and 4,216 clips never seen in training
- Sound is multi-label: spectrograms of 1 s slices every 0.5 s, several sounds at once

BANNER SHOWS

> The lab's sample photos travel, one at a time, through COLLECT → TRAIN → TEST → SHIP. At TEST each photo gets the box and score our real model gave it (for example "dolphin 66%"). (SAMPLE.)

POINTS TO COVER

- **Supervised learning:** every training photo has boxes drawn around each creature (22,718 labelled creatures across 7,637 photos); every sound clip is labelled with which sounds it contains.
- **Transfer learning:** we didn't train from zero. YOLO26 Nano (2.5 million parameters) already knew general shapes; EfficientAT mn10 was pretrained on AudioSet. We fine-tuned them on our data.
- **Fair split:** sea photos 5,312 train · 1,558 validation · 767 test. Sound clips 8,617 train · 992 validation · 4,216 test. The test set is never used for training or tuning.
- **Multi-label sound:** audio is turned into a spectrogram (a picture of sound). The model looks at 1-second slices every half second, and one slice can contain several sounds at once.

SLIDE 4 · §4 TECHNIQUES

Presenter B · The making | About 1:30 | Banner: tall strip on the right

How they predict, and how we *measure*

ON SCREEN

How they predict, and how we *measure*

- Object detection: boxes with confidence; IoU and non-max suppression remove duplicates
- Thresholds: 45% for creatures; per-sound thresholds tuned on validation, never on test
- Metrics: mAP@50, mean average precision and F1, plus a list of how each model fails
- On-device inference: exported to ONNX and run by ONNX Runtime Web in the browser

BANNER SHOWS

> A phone: the ONNX model streams in from above as particles, a photo inside gets its real box, a shield pulses, "PHOTO STAYS HERE". (SAMPLE.)

POINTS TO COVER

- **Object detection:** the model outputs boxes with a confidence. Overlapping boxes for the same animal are measured with IoU (how much two boxes overlap) and non-max suppression keeps only the best one.
- **Thresholds:** a creature is shown only if the model is at least 45% sure. Each of the 17 sounds has its own threshold, tuned on the validation set, so the test score stays honest.
- **Metrics:** mAP@50 means a box counts as correct when it overlaps the true box by at least 50%. For sound we also report F1: 0.59 second-by-second on 97 test mixes.
- **On-device:** both models are exported to ONNX and run with ONNX Runtime Web inside the browser. Nothing is uploaded.

Hand-off to Presenter C: "C will show you the two models working, and their real results."

SLIDE 5 · §5 RESULT / UI PROTOTYPE

Presenter C · The models | About 1:15 | Banner: slanted, narrower

Sea creature *detector*

ON SCREEN

MODEL 01 · UI PROTOTYPE

Sea creature *detector*

EXPECTED → BUILT ✓ 0.74 mAP@50 · 767 test photos

- Boxes 14 kinds of sea creature in a photo
- 0.82 on aquarium photos
- Fails on: seal vs sea lion, dolphin vs whale, small distant fish

BANNER SHOWS

> **Real screen captures** of the live page, one sample at a time: aquarium tank, dolphin, sea turtle, seal or sea lion (a mistake it makes), shark or fish, school of fish.

POINTS TO COVER

- This is the professor's "UI prototype" section, but it's not a mock-up: the label flips from EXPECTED to **BUILT**, and the banner is the real page.
- Score: **0.74 mAP@50** on 767 photos it never saw; **0.82** on aquarium photos.
- Be honest about failures: it mixes up seals and sea lions, dolphins and whales, and misses small, distant fish. It knows no octopus, manta ray or orca yet.
- Best and worst creatures if asked: sea turtle 0.87, starfish 0.86; puffin and fish about 0.62.

SLIDE 6 · §6 RESULT / UI PROTOTYPE

Presenter C · The models | About 1:15 | Banner: tall strip on the right

Sound *detective*

ON SCREEN

MODEL 04 · UI PROTOTYPE

Sound *detective*

EXPECTED → BUILT ✓ 0.79 mAP · 4,216 test clips

- 17 everyday sounds on a timeline
- From a recording, an upload or live listening
- Weak spots: doorbell, footsteps, car horn; timing is rough

BANNER SHOWS

The real model output for the sample "Two sounds at once": a playhead runs down the clip and bars grow for speech, engine / traffic and clapping, with their real confidences (68%, 75%, 99%...).

POINTS TO COVER

- It finds 17 everyday sounds (dog bark, siren, glass breaking, knocking...) and places them on a timeline. Three ways in: record, upload, or listen live (live mode stops by itself after 2 minutes).
- Score: **0.79 mean average precision** on 4,216 test clips.
- A design choice: the bigger mn10 model scores 0.79 at 20.5 MB; the smaller mn04 scores 0.76 at 6.5 MB. We chose accuracy.
- Honest weak spots: doorbell is the weakest sound (0.46); footsteps and car horn are also weak; timing is rough: it finds 49% of sounds within 0.5 s of when they start.

Hand-off to Presenter D: "D will tell you what we're working on now."

SLIDE 7 · EXTRA · IN PROGRESS

Presenter D · What's next | About 1:00 | Banner: band along the bottom

RAG *assistants*

ON SCREEN

IN PROGRESS

RAG *assistants*

- Answers from its own documents: retrieve the right passages, then a small language model writes the reply
- Runs in the browser: Qwen2.5-1.5B and snowflake-arctic-embed-s on WebLLM
- Two built: University of Raffel Luo and Pathum Rai District
- Tested, not complete yet: we are measuring defences against tricky prompts

BANNER SHOWS

> Questions glide along the band into a shield; past it, each one is "answered from its documents". Stamped IN PROGRESS. (SAMPLE.)

POINTS TO COVER

- RAG = retrieval-augmented generation: first search the assistant's own documents for the passages that answer the question, then a small language model writes the reply from them.
- Like the other models, it runs fully in the browser (WebLLM, Qwen2.5-1.5B for answers, snowflake-arctic-embed-s for search).
- Two assistants are built: a university helpdesk (University of Raffel Luo) and a district emergency helpdesk (Pathum Rai District).
- Status: **tested, not complete**. We are testing how well defences hold up against tricky prompts, one defence at a time and then all together. Don't quote results yet; they aren't final.

SLIDE 8 · EXTRA

Presenter D · What's next | About 0:45 | Banner: slanted, right half

What's *coming*

ON SCREEN

What's *coming*

- RAG testing: finish measuring each defence against tricky prompts on both assistants, then all of them combined
- A private document assistant as the next RAG test target
- More sea creatures: octopus, manta ray, orca, lionfish
- Sound detective v2: better timing, more sounds
- A speech translator and text models

BANNER SHOWS

> New specimen plates (Octopus, Manta ray, Orca, Lionfish) fly in from the depth of the swarm. (SAMPLE.)

POINTS TO COVER

- First priority: finish the RAG testing, every defence alone, then all combined, on both assistants.
- Next RAG target: a private document assistant.
- Widen the sea detector to the creatures it can't find yet; Sound detective v2 with better timing (training data with timestamps) and more sounds.
- Later: a speech translator and text models.

Hand-off to Presenter E: "E will wrap up."

SLIDE 9 · §7 CONCLUSION & REFERENCES

Presenter E · Conclusion | About 1:00 | Banner: floating monitor + QR code

What we *built*

ON SCREEN

What we *built*

- 2 working models, running on your device
- 2 RAG assistants built, still being tested
- Every number is measured; every report card shows how it fails
- Data: Open Images V7 (CC BY 2.0), Aquarium Combined (Roboflow, CC BY 4.0), FSD50K and Freesound (CC0 / CC BY)
- Models: Ultralytics YOLO26 (AGPL-3.0), EfficientAT (MIT), Qwen2.5, snowflake-arctic-embed
- Tools: ONNX Runtime Web, WebLLM, Remotion
- Music: "Tech Circuit Data Stream" by Alex Morgan (Pixabay licence)

`+ a large QR code and the site address: "Try it live"`

BANNER SHOWS

> A slow fly-through of the real home page of the live site.

POINTS TO COVER

- Summary: two working models that run on your device, and two RAG assistants that are built and still being tested.
- What makes the lab different: every number is measured, and every model shows how it fails.
- Read the references briefly: where the data, models and music come from, and their licences.
- Invite everyone to scan the QR code and try the models on their phones. Then thank the audience and ask for questions.